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A Case Study on the Causes and Consequences of Road Traffic Accidents in Saptari District

Vidyanand Yadav[1], **Ambira Pradhan**[2]

[1] *Lecturer in Statistics, Shree M. B. M. Campus, Rajbiraj, Saptari, Nepal.*
[2] *Orchid International College, Gausala, Kathmandu, Nepal*

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ABSTRACT

Background:: Road traffic accidents are events occurring on roads involving at least one moving vehicle and may result in injury, death, property loss, and substantial social and economic consequences. This study investigated the causes and consequences of road traffic accidents in Saptari District, Nepal.
Methods:: A quantitative and descriptive study design was employed, combining primary and secondary data. Primary information was collected from 60 vehicle drivers from Ward No. 09 of Rajbiraj Municipality, while secondary information was obtained from the District Traffic Police Office, Saptari, covering the period 2019–2024 AD. The analysis considered vehicle accidents, types of vehicles involved, serious injuries, human deaths, animal deaths, and the gender and age distribution of casualties. Descriptive statistics, percentages, graphical presentations, chi-square goodness-of-fit test, and linear trend analysis were used.
Results:: Motorcycles constituted the largest proportion of vehicles involved in recorded road traffic accidents, while alcohol consumption while driving was the leading reported cause among surveyed drivers. Young adults were particularly affected by road traffic accidents. The highest annual number of vehicle accidents was reported in 2022, whereas the highest numbers of deaths and serious injuries occurred in 2021. Based on the aggregate annual death series, the fitted linear trend estimated approximately 90 deaths in 2026 and 98 deaths in 2030.
Conclusion:: The findings indicate a substantial road traffic accident burden in Saptari District and highlight the importance of addressing modifiable human and road-related risk factors. Strengthening road-safety awareness, enforcing traffic regulations, improving road conditions, promoting responsible driving practices, and implementing targeted safety interventions for motorcycle users are recommended to reduce road traffic injuries and fatalities.

Keywords: Road traffic accidents; road safety; over-speeding; drunk driving; motorcycle accidents; Saptari; Nepal.

Correspondence: Mr. Vidyanand Yadav, Shree M. B. M. Campus, Rajbiraj, Saptari, Nepal. E-mail: bidyanandpersonal@gmail.com.

INTRODUCTION

An event which occurs on a public road, way, or street resulting in one or more persons being injured or killed, where at least one moving vehicle is involved, is known as a road accident. A road accident may involve collisions between vehicles, between a vehicle and an animal, between a vehicle and a geographical obstacle, or between vehicles and pedestrians.

Road traffic accidents are a major public-health and socioeconomic concern. The World Health Organization has reported approximately 1.19 million road-traffic deaths annually worldwide, with thousands of people dying every day from road crashes [1]. In the South-East Asia region, road traffic deaths represent a substantial proportion of the global burden [2].

Young people are particularly vulnerable to road traffic injuries, and pedestrians, cyclists, motorcyclists constitute important groups among road-traffic victims. In Nepal, the number of motor vehicles has increased rapidly, creating additional challenges for road safety. Nepal Police periodically report substantial numbers of deaths & injuries associated with road traffic crashes [3]. According to Nepal Police statistics, 2,540 persons died and 5,646 were seriously injured in 2019/20 AD; 2,251 died and 4,615 were injured in 2020/21 AD; 2,500 died and 6,448 were injured in 2021/22 AD; 2,823 died and 7,282 were injured in 2022/23 AD; and 2,363 died and 5,738 were injured in 2023/24 AD.

Road traffic accidents result not only in loss of life but also in physical suffering, psychological distress, reduced productivity, and economic losses for affected families and communities. The economic burden of road traffic injuries is particularly important in developing countries [4].

The present study examines the trend, indicators, comparison, and characteristics of road traffic accidents in Saptari District, Nepal. It focuses on the significance of road safety and the need to reduce the number of accidents and their consequences. The study also considers driver behaviour, road infrastructure, vehicle conditions, traffic rules, and public awareness.

Despite the continuing burden of road traffic accidents in Nepal, there is a need for a more detailed understanding of the pattern and characteristics of road traffic accidents at the district level. In particular, evidence on the variation of accidents, their associated indicators, and the characteristics of those involved is important for identifying areas requiring greater attention to road safety. However, district-specific analysis can provide more useful insights for understanding the local context and supporting appropriate preventive measures.

Therefore, the present study examines the trends, indicators, comparisons, and characteristics of road traffic accidents in Saptari District, Nepal, with the aim of providing evidence that may contribute to improved road safety and the reduction of road traffic accidents and their consequences.

METHODS

The study was conducted in Saptari District, Nepal. The population considered in the study included persons injured or killed in road traffic accidents, animals involved in accidents, vehicle drivers, and vehicles associated with traffic accidents.

The study also considered road and market conditions, traffic behaviour, and public awareness related to road use. Particular attention was given to accidents associated with careless driving, vehicle conditions, road conditions, and the movement of people and animals in market areas.

This study employed a quantitative and descriptive research design. A multistage random sampling approach was used for the primary survey.

First, Saptari District was selected as the study area. In the second stage, Rajbiraj Municipality was selected by the lottery method from the municipalities of Saptari District. In the third stage, 60 vehicle drivers from Ward No. 09 of Rajbiraj Municipality were selected for the primary survey.

Primary data were collected from 60 vehicle drivers from Ward No. 09 of Rajbiraj Municipality using structured questions regarding their experience and perceptions of road traffic accidents.

Secondary quantitative information on road traffic accidents was obtained from the District Traffic Police Office, Saptari. The secondary dataset covered the period from 2019 AD to 2024 AD and included vehicle accidents, serious injuries, deaths, animal deaths, gender, age, and vehicle categories.

The factors considered in relation to road traffic accidents included:

- different types of vehicles;
- poor road conditions;
- crowding in market areas;
- animals moving on roads; and
- driver-related behaviours and practices.

The sample-size expression used was [5]

$$n = \frac{Z^2 pq}{e^2}, \tag{1}$$

where Z is the standard normal deviate corresponding to the desired confidence level, p is the assumed prevalence, $q = 1 - p$, and e is the permissible margin of error. Assuming a 95% confidence level ($Z = 1.96$), an expected prevalence of 80% ($p = 0.80$), and a permissible error of 10% ($e = 0.10$), the calculated sample size was

$$n = \frac{(1.96)^2(0.80)(0.20)}{(0.10)^2} = 61.47.$$

The primary survey included 60 respondents.

The data were analyzed using descriptive and inferential statistical techniques. Descriptive statistics, including frequency, percentage, mean, and standard deviation, were used to summarize the characteristics and patterns of road traffic accidents [6]. Graphical representations, including pie charts and trend-line plots, were used to illustrate the distribution and temporal pattern of accidents and deaths.

A chi-square goodness-of-fit test was applied to assess whether the annual number of deaths differed significantly from a uniform distribution across the six study years. Linear trend analysis was used to examine the temporal trend in the annual number of deaths, with the fitted trend represented by $Y_c = a + bx$, where Y_c denotes the estimated number of deaths and x represents the coded time variable. Statistical significance was assessed at the 5% level.

RESULTS

The primary survey included 60 vehicle drivers from Ward No. 09 of Rajbiraj Municipality. Respondents aged 20–24 years constituted the largest age group (41.67%), followed by those aged 25–29 years (25.00%). Respondents below 20 years and those aged 30 years or older each accounted for 16.67%.

Table 1. Age distribution and reported involvement in road accidents among survey respondents

Age group			Accident involvement		
Category	n	%	Category	n	%
Below 20	10	16.67	Yes	35	58.33
20–24	25	41.67	No	10	16.67
25–29	15	25.00	Rarely	15	25.00
30 and above	10	16.67			
Total	60	100.00	Total	60	100.00

Note: Percentages may not sum to exactly 100 due to rounding.

Of the 60 respondents, 35 (58.33%) reported having been involved in a road traffic accident, 15 (25.00%) reported rare involvement, and 10 (16.67%) reported no involvement. These findings describe respondents' reported experiences and should not be interpreted as an estimate of district-level accident incidence.

Respondents identified several potential contributors to road traffic accidents, including excessive speed, narrow and curved roads, overtaking, alcohol consumption while driving, adverse weather, old vehicle conditions, overloading, and improper driving skills. Alcohol consumption while driving was the most frequently reported contributor (25.00%), followed by excessive-speed driving (20.00%), overtaking (16.67%), and improper driving skills (13.33%). Because the survey was based on respondents' reports, these findings represent perceived or reported contributors rather than causal estimates.

The secondary administrative dataset recorded 7,895 vehicle accidents during 2019–2024. The annual number increased from 250 in 2019 to 2,156 in 2022, which represented 27.31% of all recorded vehicle accidents, and subsequently declined to 1,523 in 2024. The mean annual number of recorded vehicle accidents was 1,315.83 (SD = 707.15).

The largest number of vehicle accidents was recorded in 2022, whereas the largest numbers of serious injuries and deaths were recorded in 2021. Across the six-year period, the administrative records contained 5,649 serious injuries and 482 deaths. The corresponding annual means were 941.50 serious injuries and 80.33 deaths. Animal deaths were comparatively infrequent, with 25 recorded over the study period; the highest annual number was observed in 2021 (10, 40.00% of the six-year total).

The administrative records also contained sex-disaggregated information for a subset of the aggregate records. The sex-specific totals therefore do not represent the complete annual totals in Table 2.

Among records with available sex information, males constituted the majority of both deaths and serious injuries in every year. Male deaths accounted for 80.60% of the sex-recorded deaths overall, while males accounted for 81.16% of sex-recorded serious injuries. The highest proportion of male deaths was 87.21% in 2022; the highest proportion of male serious injuries was 85.71%, observed in 2020 and 2024.

Vehicle-category data showed that motorcycles were the most frequently recorded vehicle category in every year. Their share increased from 44.98% in 2019 to 64.89% in 2024, reaching 61.77% in 2022. The denominators in this table are vehicle involvement records and therefore differ from the number of vehicle accidents reported in Table 2; a single accident may involve more than one vehicle.

The age-disaggregated records comprised 1,547 observations with available age information. The 25–34 years group had the largest number of recorded deaths and serious injuries (445), followed by the 15–24 years group (376). Together, these two age groups accounted for 821 observations (53.07%) in the age-disaggregated dataset.

The 25–34 years group contributed 28.77% of the age-disaggregated records, while the 15–24 years group contributed 24.31%. These figures indicate a substantial burden among young and working-age groups, although the absence of complete age information precludes interpretation as population-based age-specific rates.

The graphical representation is consistent with the tabulated data, showing the largest annual share of vehicle accidents in 2022 and the largest annual share of deaths in 2021.

A chi-square goodness-of-fit test was used to assess whether the aggregate annual number of deaths was consistent with a uniform distribution across the six study years.

The test was based on the aggregate death series in Table 2, not on the incomplete sex-disaggregated series. With 482 observed deaths over six years, the expected number per year under H_0 was

$$E_i = \frac{482}{6} = 80.33.$$

Table 6. Chi-square goodness-of-fit test for the annual distribution of deaths

O_i	E_i	$(O_i - E_i)^2$	$\frac{(O_i-E_i)^2}{E_i}$
69	80.33	128.44	1.60
70	80.33	106.78	1.33
97	80.33	277.78	3.46
89	80.33	75.11	0.94
75	80.33	28.44	0.35
82	80.33	2.78	0.03
482	482.00		7.71

The calculated chi square value was $\chi^2 = 7.71$, with $df = 5$, with corresponding p-value was approximately 0.173. There is no any statistically significant departure from a uniform distribution of annual deaths over the six-year period. This finding does not imply that the annual death counts were identical or that there was no temporal variation.

The fitted linear trend was defined as

$$Y_c = a + bx, \tag{2}$$

where Y_c denotes the fitted number of deaths. The fitted trend equation was found as

$$Y_c = 80.33 + 2.057x. \tag{3}$$

For 2026, $x = 4.5$, giving

$$Y_c = 80.33 + 2.057(4.5) = 89.59 \approx 90.$$

For 2030, $x = 8.5$, giving

$$Y_c = 80.33 + 2.057(8.5) = 97.82 \approx 98.$$

These values are simple linear extrapolations rather than formal forecasts and should be interpreted cautiously because the model is based on only six annual observations and does not account for changes in traffic exposure, vehicle ownership, enforcement, road infrastructure, or emergency care.

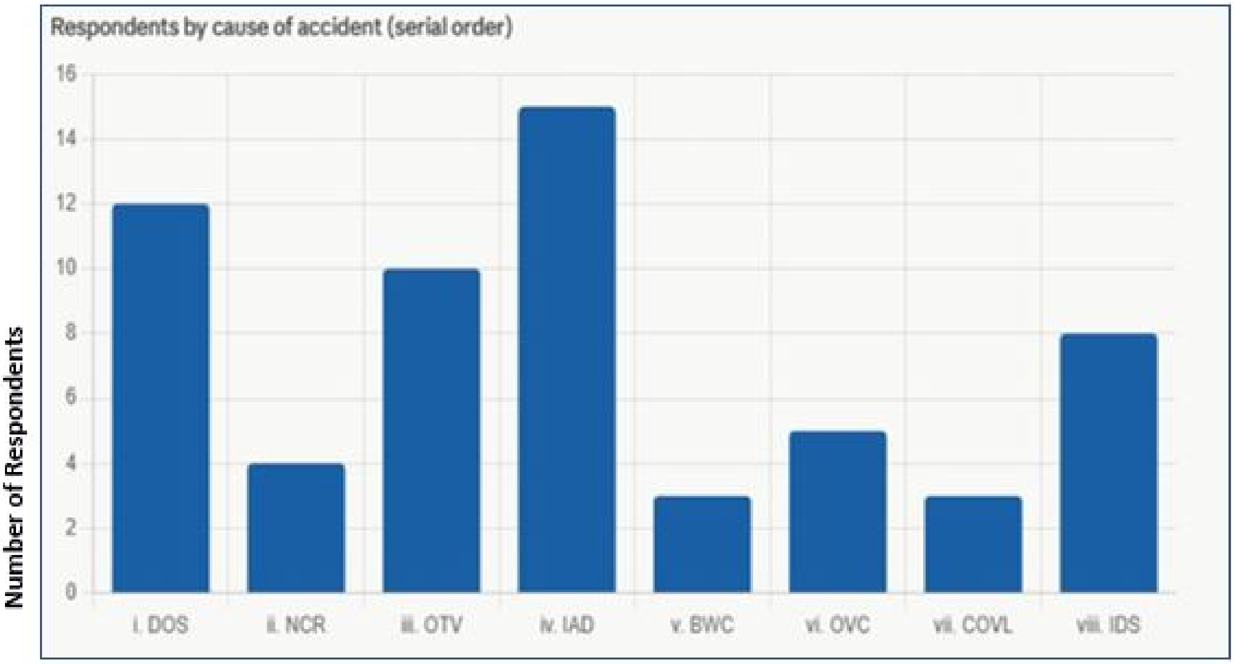

Figure 1. Reported contributors to road traffic accidents among surveyed drivers

Table 2. Annual distribution of vehicle accidents and reported consequences, 2019–2024

Year	Vehicle accidents, n (%)	Serious injuries, n (%)	Deaths, n (%)	Animal deaths, n (%)
2019	250 (3.17)	140 (2.48)	69 (14.32)	2 (8.00)
2020	469 (5.94)	253 (4.48)	70 (14.52)	4 (16.00)
2021	1,864 (23.61)	1,558 (27.58)	97 (20.12)	10 (40.00)
2022	2,156 (27.31)	1,450 (25.69)	89 (18.46)	3 (12.00)
2023	1,633 (20.68)	1,197 (21.19)	75 (15.56)	3 (12.00)
2024	1,523 (19.29)	1,051 (18.61)	82 (17.01)	3 (12.00)
Total	7,895 (100.00)	5,649 (100.00)	482 (100.00)	25 (100.00)

DISCUSSION

This study provides a district-level assessment of road traffic accidents in Saptari District by combining a primary survey of vehicle drivers with administrative records from the District Traffic Police Office. Three findings are particularly important. First, alcohol consumption while driving was the most frequently reported contributor in the primary survey, followed by over-speeding, overtaking, and improper driving skills. Second, motorcycles accounted for the largest proportion of vehicles recorded in the administrative data. Third, the administrative records documented a substantial burden of mortality and serious injury, with the highest annual numbers of both deaths and serious injuries occurring in 2021. Collectively, these findings indicate that road-safety interventions in Saptari should address modifiable driving behaviours alongside vehicle and road-environmental factors.

The prominence of alcohol consumption, speeding, overtaking, and improper driving skills in the primary survey is consistent with the established importance of human behaviour in road traffic injury prevention [7–9]. These findings should nevertheless be interpreted as respondents' reported perceptions rather than independently demonstrated causal effects. The cross-sectional design and self-reported nature of the survey do not allow the relative contribution of individual behaviours to crash occurrence to be established. Even so, the concentration of responses around potentially modifiable behaviours provides a practical basis for prevention. Interventions combining enforcement with driver education may be particularly relevant, with emphasis on impaired driving, speed management, and safe overtaking practices.

Motorcycles were the dominant vehicle category in the administrative records, accounting for 61.77% of recorded vehicles in 2022. This finding is important for local road-safety planning because motorcycle users have comparatively limited physical protection during collisions. However, the present analysis describes the distribution of vehicles involved in reported accidents rather than crash risk per vehicle or per unit of travel. Therefore, the high proportion of motorcycles should not be interpreted as evidence that motorcycles necessarily have a higher individual-level crash risk than other vehicle categories. Such an inference would require exposure de-

Table 3. Sex distribution of recorded deaths and serious injuries with available sex information

Year	Deaths		Serious injuries	
	Male, n (%)	Female, n (%)	Male, n (%)	Female, n (%)
2019	51 (73.91)	18 (26.09)	55 (75.34)	18 (24.66)
2020	55 (84.62)	10 (15.38)	108 (85.71)	18 (14.29)
2021	70 (80.46)	17 (19.54)	310 (80.52)	75 (19.48)
2022	75 (87.21)	11 (12.79)	170 (79.07)	45 (20.93)
2023	53 (75.71)	17 (24.29)	159 (79.10)	42 (20.90)
2024	45 (80.36)	11 (19.64)	180 (85.71)	30 (14.29)
Total	349 (80.60)	84 (19.40)	982 (81.16)	228 (18.84)

Table 4. Distribution of vehicle categories involved in reported road accidents, 2019–2024

Vehicle category	2019	2020	2021	2022	2023	2024
Truck	9 (6.04)	21 (7.58)	54 (4.78)	53 (4.08)	31 (3.21)	24 (2.71)
Tipper	2 (1.34)	8 (2.89)	10 (0.88)	7 (0.54)	4 (0.41)	9 (1.01)
Bus	20 (13.42)	20 (7.22)	25 (2.21)	79 (6.08)	58 (6.01)	37 (4.18)
Car/Jeep	17 (11.41)	25 (9.03)	102 (9.00)	86 (6.62)	48 (4.97)	52 (5.89)
Tractor	14 (9.40)	18 (6.57)	48 (4.24)	56 (4.31)	40 (4.15)	37 (4.18)
Tempu	3 (2.01)	4 (1.44)	56 (4.94)	59 (4.54)	66 (6.84)	52 (5.89)
Motorcycle	67 (44.98)	145 (52.35)	671 (59.22)	803 (61.77)	587 (60.83)	574 (64.89)
Others	17 (11.41)	36 (12.99)	167 (14.74)	157 (12.08)	131 (13.58)	100 (11.30)
Total involvement records	149	277	1,133	1,300	965	885

nominators, such as the number of registered vehicles or vehicle-kilometres travelled. Nevertheless, the observed pattern provides a clear rationale for giving motorcycle safety substantial attention within local prevention programmes.

The administrative dataset recorded 482 deaths and 5,649 serious injuries during 2019–2024. The highest annual numbers of deaths and serious injuries were recorded in 2021, whereas the largest number of vehicle accidents was recorded in 2022. The difference between the peak in accident frequency and the peak in mortality illustrates that the number of reported crashes alone does not fully describe road-traffic severity. Crash circumstances, road conditions, traffic exposure, emergency response, and the severity of individual events may influence mortality independently of the total number of reported accidents.Adverse weather and environmental conditions may also influence road traffic crash occurrence and severity [10]. The substantial burden observed in Saptari is consistent with the broader road-traffic injury burden reported in Nepal [3, 11].

The available sex-disaggregated records showed a marked predominance of males among both deaths and serious injuries. Male deaths accounted for more than three-quarters of the recorded deaths in each year of the available series, and a similar pattern was observed for serious injuries. This is broadly consistent with previous evidence of higher road-traffic mortality among males [11, 12]. Differences in mobility, occupational exposure, vehicle use, and engagement in higher-risk driving behaviours may contribute to this pattern. However, the present study did not collect sufficient information to evaluate these mechanisms, and the observed sex distribution should therefore be interpreted as a descriptive finding rather than evidence of a specific causal pathway.

The age-disaggregated records indicated that the 25–34 years group had the largest number of reported deaths and serious injuries, followed by the 15–24 years group. The concentration of reported events among younger and working-age adults has important social and economic implications because injury and premature mortality in these groups can affect employment [13, 14], household income, and family responsibilities. The economic consequences of road traffic injuries extend beyond immediate medical expenditure and may include productivity losses and long-term disability [4]. Because the age-disaggregated records represent observations for which age information was available, however, these findings should not be interpreted as population-based age-specific risk estimates.

The chi-square goodness-of-fit analysis produced $\chi^2 = 7.71$ with 5 degrees of freedom and $p \approx 0.173$, providing insufficient evidence to reject the specified uniformity hypothesis at the 5% significance level. This result should not be interpreted as evidence that mortality was constant across the study period; rather, it indicates that the available six-year series did not provide sufficient statistical evidence of a departure from the hypothesised uniform distribution. The short time series also limits the power and interpretability of formal temporal analyses [15]. Similarly, the linear trend produced extrapolated values of approximately 90 deaths in 2026 and 98

Table 5. Age distribution of recorded deaths and serious injuries with available age information

Age group	2019	2020	2021	2022	2023	2024	Total
0–14	13	17	47	25	25	15	142
15–24	27	32	129	87	60	41	376
25–34	38	50	139	98	72	48	445
35–44	28	36	71	36	43	29	243
45–54	18	34	66	27	30	19	194
55–64	13	13	11	16	28	10	91
65+	5	9	9	12	13	8	56
Total	142	191	472	301	271	170	1,547

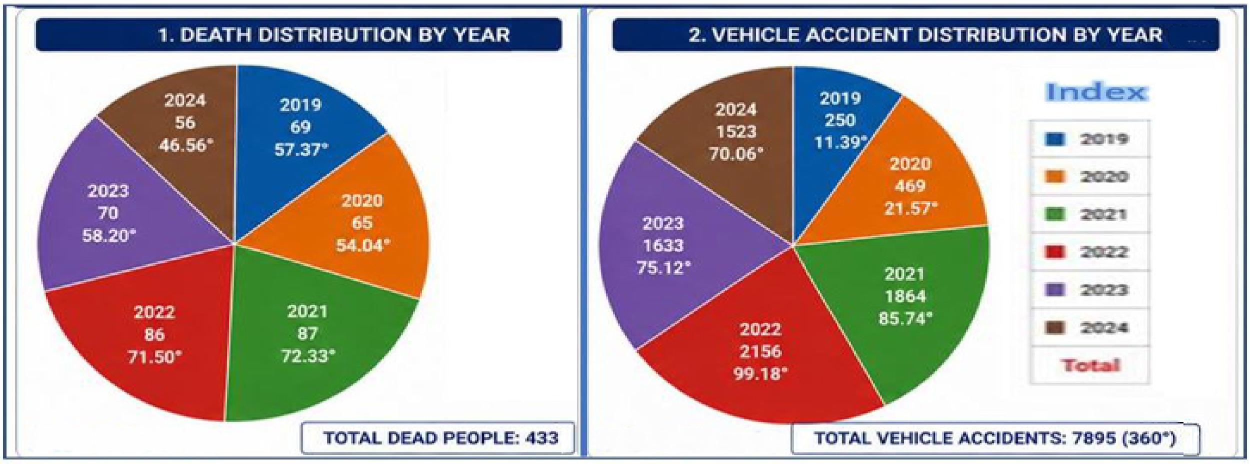

Figure 2. Graphical representation of annual vehicle accidents and deaths, 2019–2024

deaths in 2030. These values should be regarded as simple descriptive projections rather than formal forecasts, because they are based on only six annual observations and do not incorporate changes in traffic exposure, vehicle ownership, road infrastructure, enforcement, emergency care, or other determinants of mortality.

From a public-health and policy perspective, the findings support a multidimensional road-safety strategy in Saptari District. Enforcement against impaired and excessive-speed driving should be accompanied by evidence-informed driver education, motorcycle-focused safety programmes, helmet compliance, safer overtaking practices, and regular vehicle inspection. Local authorities should also consider engineering measures in locations characterised by narrow or curved roads, high pedestrian activity, or interaction between vehicles and animals. Where possible, interventions should be guided by geographically disaggregated crash data so that limited resources can be directed toward locations and population groups with the greatest burden. Importantly, road-safety prevention should not rely solely on individual behaviour; safer infrastructure, effective enforcement, and timely post-crash care are complementary components of an effective road safety system.

CONCLUSION

This study demonstrates a substantial burden of road traffic accidents in Saptari District, with motorcycles accounting for the largest proportion of vehicles recorded and alcohol consumption while driving, over-speeding, overtaking, and improper driving skills emerging as the most frequently reported behavioural contributors. During 2019–2024, the administrative records documented 482 deaths and 5,649 serious injuries, with the highest annual numbers of both outcomes occurring in 2021. The available sex-disaggregated records showed a predominance of males, while the age-disaggregated records indicated a substantial burden among young and working-age adults. Although the annual death series did not show a statistically significant departure from the specified uniformity hypothesis, the continuing occurrence of deaths and serious injuries highlights the need for sustained prevention.

The findings support an integrated road-safety strategy combining enforcement of impaired- and excessive-speed driving regulations with motorcycle-focused interventions, driver education, safer road design, vehicle safety measures, and strengthened post-crash response. The findings should not be interpreted as causal estimates of individual risk because the primary survey was small and the administrative data lacked exposure denominators. Future research using larger representative samples, linked administrative and clinical records, and exposure-adjusted measures is needed to identify modifiable determinants of crash occurrence and severity. Such evidence would provide a stronger basis for prioritising locally appropriate interventions and for monitoring progress in reducing road traffic mortality and injury in Saptari District.

A strength of this study is the use of complementary primary and administrative data, allowing reported driver-level contributors to be considered alongside annual patterns in road traffic events and their consequences. The analysis also brings together information on vehicle involvement, mortality, serious injury, animal deaths, sex, and age, providing a multidimensional description of the local road-safety burden.

The study is limited by the small, single-ward driver sample and reliance on self-reported data, which may introduce selection, recall, and reporting bias. Administrative records from a single district may not capture all crashes, while missing demographic and exposure data limited subgroup analysis and calculation of population- or vehicle-based crash rates. The six-year series also restricts the reliability of long-term trend forecasting.

The findings support an integrated road-safety response in Saptari District. Enforcement should prioritise impaired and excessive-speed driving and should be accompanied by sustained driver education, safe overtaking practices, and appropriate use of helmets and seat belts. Given the consistently high proportion of motorcycle involvement, motorcycle-specific safety interventions should receive particular attention. Local authorities should also consider engineering improvements at locations characterised by narrow or curved roads, high pedestrian activity, and interaction between vehicles and animals. Strengthening routine linkage of police, hospital, ambulance, and mortality records would improve surveillance and allow future analyses to incorporate exposure denominators and measures of crash severity. These measures would provide a stronger evidence base for targeting limited road-safety resources and evaluating intervention effectiveness.

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